Chapter 5: Map Reduce Based Machine Learning

❖ Topic covered:

- Module content: K-Means, PLANET, Parallel SVM,
- Association Rule Mining in MapReduce
- Inverted Index,Page Ranking
- Expectation Maximization
- Bayesian Networks

❖ What is MapReduce in machine learning?

- MapReduce is a framework for processing parallelizable problemsacross large datasets using a large number of computers (nodes),collectively referred to as a cluster (if all nodes are on the same localnetwork and use similar hardware) or a grid (if the nodes are sharedacross geographically and administrative.

1. MapReduce Overview:

- MapReduce breaks down large data sets into smaller, manageable chunks, which are then processed concurrently.
- It consists of two essential functions:
  - Map Function: Manipulates input data and produces a collection of key-value pairs.
  - Reduce Function: Aggregates and combines the intermediate results generated by the Map function.

2. How MapReduce Works:

  - A client submits a MapReduce job to the Hadoop MapReduce Master.
  - The master divides the job into equivalent job-parts.
  - Map and Reduce tasks execute the program logic specified by developers.
  - Input data is fed to the Map task, which generates intermediate key-value pairs.
  - These pairs are then processed by the Reducer, and the final output is stored in the Hadoop Distributed File System (HDFS).

- Multiple Map and Reduce tasks can run concurrently, optimizing performance and minimizing overhead.

3. Use Cases:
   - Big Data Analytics: MapReduce enables efficient processing of large-scale data for tasks like data mining, log analysis, and recommendation systems.
   - Machine Learning: While not a machine learning algorithm itself, MapReduce supports preprocessing steps (feature extraction, data transformation) required for ML models.
   - Graph Processing: MapReduce is used for graph algorithms like PageRank and community detection.

4. Importance:
   - Scalability: MapReduce scales horizontally by distributing computation across nodes, making it suitable for massive datasets.
   - Fault Tolerance: If a node fails, MapReduce redistributes tasks to other nodes, ensuring robustness.
   - Parallelism: Parallel processing improves performance by utilizing available resources efficiently.

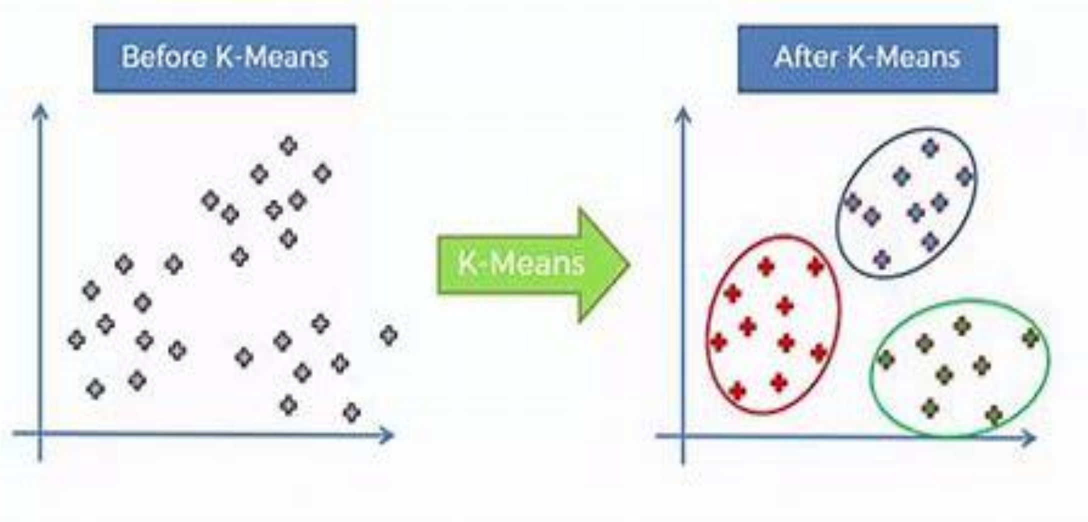
❖ **K-Means:**

- K-Means Clustering is an unsupervised learning algorithm that isused to solve the clustering problems in machine learning or data science. In this topic, we will learn what is K-means clustering algorithm, how the algorithm works, along with the Python implementation of k-means clustering.

❖ **What is K-Means Algorithm?**

- K-Means Clustering is an Unsupervised Learning algorithm, which groups the unlabeled dataset into different clusters. Here K defines the number of predefined clusters that need to be created in the process, as if K=2, there will be two clusters, and for K=3, there will be three clusters, and so on.
- It allows us to cluster the data into different groups and a convenient way to discover the categories of groups in the unlabeled dataset on its own without the need for any training.
- It is a centroid-based algorithm, where each cluster is associated with a centroid. The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding clusters.

- The algorithm takes the unlabeled dataset as input, divides the dataset into k-number of clusters, and repeats the process until it does not find the best clusters. The value of k should be predetermined in this algorithm.
- The k-means clustering algorithm mainly performs two tasks:
- Determines the best value for K center points or centroids by an iterative process.
- Assigns each data point to its closest k-center. Those data points which are near to the particular k-center, create a cluster.
- •Hence each cluster has datapoints with some commonalities, and it is away
- from other clusters.
- •The below diagram explains the working of the K-means Clustering Algorithm:



❖ PLANET:

➢ Big data analytics has a significant role in understanding and addressing environmental challenges. Let's explore the intersection of big data, sustainability, and our planet:

- Data-Driven Sustainability:
  - Analytics applications range from descriptive and diagnostic analytics (understanding what happened and why) to predictive and prescriptive analytics (predicting and influencing future outcomes).

- While businesses operate within specific domains, sustainability is an externality for most business models. Measuring and promoting sustainability often falls on governments, NGOs, and inter-governmental organizations.
- The Sustainable Development Goals (SDGs), a set of 17 global goals established by the United Nations, address broad and ambitious objectives such as poverty reduction, sustainable cities, and climate action.

- Challenges and Opportunities:
  - Measuring Progress: Organizations face challenges in measuring progress toward sustainability goals. Unlike businesses, these organizations lack well-defined stakeholders and responsibilities.
  - Resource Consumption vs. Efficiency: Big data can help measure sustainability, but it also consumes resources. Balancing improved efficiency with increased resource consumption is critical.
  - Environmental Impact: We must consider the net effect of analytics applications on the environment as a whole. How do we minimize negative impacts while maximizing positive outcomes?

- Satellite Imagery Analytics:
  - Companies like Planet use deep learning and computer vision to transform daily satellite imagery into actionable insights.
  - Planet Analytics detects and classifies objects, identifies geographic features, and monitors changes globally. It provides data feeds in formats like GeoJSON and GeoTIFF.

- Giant Planets and Big Data:
  - Researchers have developed techniques like PlanetNet, which identifies and maps components and features in turbulent regions of planets like Saturn. Deep learning reveals insights into atmospheric processes.

❖ Parallel SVM:

➢ What is parallel SVM?

- The new parallel incremental Support Vector Machine (SVM) algorithm aims at classifying very large datasets on graphics processing units (GPUs).

➢ **There are two different types of SVMs, each used for different things:**
- Simple SVM: Typically used for linear regression and classification problems.
- Kernel SVM: Has more flexibility for non-linear data because you can add more features to fit a hyper plane instead of a two-dimensional space.

❖ **Association Rule Mining in MapReduce:**

➢ Association rule mining is a powerful technique used to discover interesting relationships or associations between items in large datasets. In the context of MapReduce, association rule mining can be efficiently performed using parallel processing. Let's explore how this works:

1. The Apriori Algorithm:

   The Apriori algorithm is a well-known method for association rule mining. It identifies frequent patterns (itemsets) in data.
   The process involves iteratively finding k-itemsets based on (k-1)-itemsets. Frequent 1-itemsets are initially identified by scanning the database and accumulating their counts. These itemsets meet a minimum support threshold.

2. Challenges in Big Data Environment:

   Traditional association rule mining algorithms, like Apriori, mostly focus on mining positive association rules. These rules indicate positive correlations between items (e.g., market basket analysis).
   However, in today's big data environment, we need to extend association rule mining to handle both positive and negative association rules.

3. Hadoop's MapReduce Environment:

   Using Hadoop's distributed and parallel MapReduce environment, we can mine both positive and negative association rules efficiently.

4. Applications:

Positive association rule mining is widely used in market basket analysis, biological datasets, web logs, fraud detection, and more. Negative association rules can provide valuable insights in various domains.

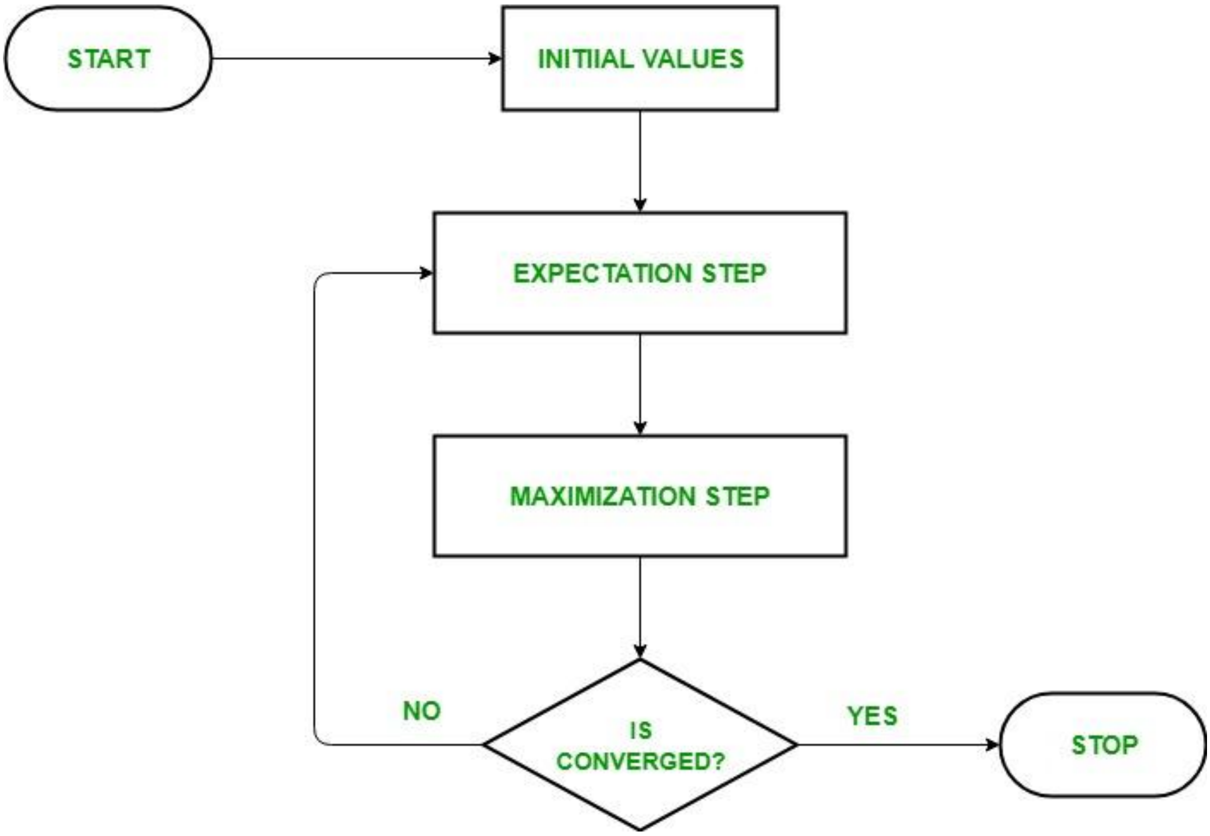
❖ **Inverted Index , Page Ranking:**

➢ Inverted Index:
- An inverted index is a data structure used in information retrieval systems to efficiently retrieve documents or web pages containing specific terms.

➢ In an inverted index:
- The index is organized by terms (words).
- Each term points to a list of documents or web pages that contain that term.

❖ **Page Ranking:**
- PageRank is an algorithm developed by Google to rank web pages in search engine results.
- It assigns a numerical value (PageRank score) to each page based on:
  - The number and quality of links pointing to that page.
  - The importance of the linking pages.
- Higher PageRank indicates greater authority or relevance.
- Google combines PageRank with other relevance scores to rank pages for specific queries.

❖ **Expectation Maximization:**



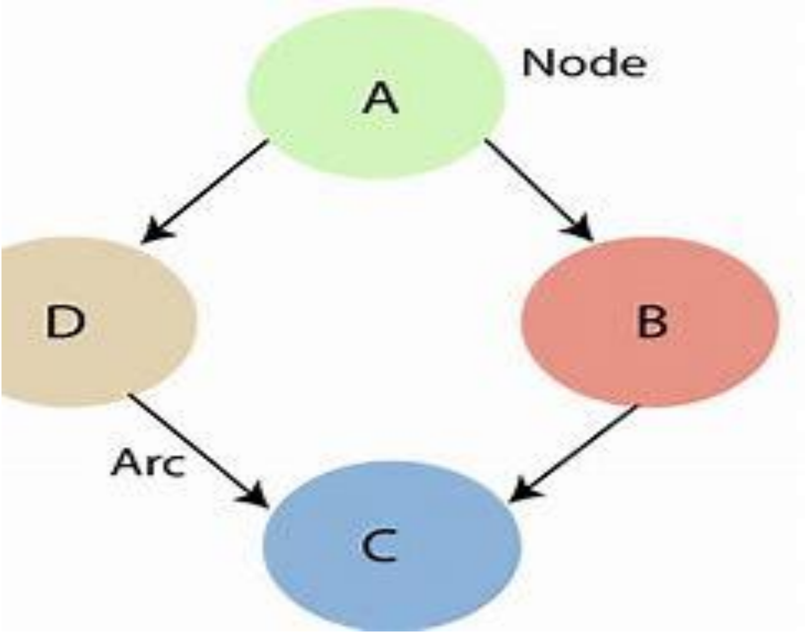
- The Expectation-Maximization (EM) algorithm is a powerful technique for parameter estimation in models involving latent variables and missing data.

- **How EM Works:**
- Initialization:
  - Start with an initial guess for model parameters.
- Expectation Step (E-Step):
  - Use the observed data to estimate or guess the values of missing data (latent variables).
  - Obtain complete data by imputing missing values.
- Maximization Step (M-Step):
  - Use the complete data from the E-step to update the model parameters.
  - Maximize the likelihood function through parameter updates.
- Iteration:
  - Repeat the E-step and M-step until convergence.

❖ **Bayesian Networks:**

- ❖ Bayesian belief network is key computer technology for dealing with probabilistic events and to solve a problem which has uncertainty. We can define a Bayesian network as:
- ❖ "A Bayesian network is a probabilistic graphical model which represents a set of variables and their conditional dependencies using a directed acyclic graph."
- ❖ It is also called a **Bayes network, belief network, decision network**, or **Bayesian model**.
- ❖ Bayesian networks are probabilistic, because these networks are built from a **probability distribution**, and also use probability theory for prediction and anomaly detection.

➢ **A Bayesian network graph is made up of nodes and Arcs (directed links), where:**



- ○ Each node corresponds to the random variables, and a variable can be continuous or discrete.
- ○ Arc or directed arrows represent the causal relationship or conditional probabilities between random variables. These directed links or arrows connect the pair of nodes in the graph.
- ○ These links represent that one node directly influence the other node, and if there is no directed link that means that nodes are independent with each other.